WORKSHEET 1 · NO CODE NEEDED

How far can you turn the wall before it goes blind?

Name _____ Team _____ Date _____

Set up. Stand a flat board exactly **20 cm** from the sensor, square on. Put the brick on Port View so you can see the reading. You will turn the board a little at a time and write down what happens.

PREDICT FIRST — BEFORE YOU TURN ANYTHING

At how many degrees do you think it will stop working? _____ degrees

Why do you think that? _____

Turn the board five degrees at a time

BOARD TURNED	WHAT THE SCREEN SAYS	DID ANYTHING COME BACK?
0°		
5°		
10°		
15°		
20°		
25°		
30°		
35°		

WHAT YOU FOUND

It stopped working at about _____ degrees.

My prediction was _____ degrees.

Before it failed completely, the number was getting _____.

THE QUESTION THAT MATTERS

Did the reading get **slowly worse**, or did it work fine and then stop **all at once**?

_____

WORKSHEET 2 · NO CODE NEEDED

What sends the click back, and what does not?

Name _____ Team _____ Date _____

Set up. Hold each surface **20 cm** from the sensor, square on, and read the screen. Same distance every time — the only thing you are allowed to change is **what the surface is made of**.

PREDICT FIRST

Which one do you think the robot will lose completely? _____

Will black and white read the same, or different? _____

Test each one at 20 cm

SURFACE	YOU PREDICTED	WHAT IT READ	CAME BACK?
A hard flat board or book			
A cushion, coat or fleece			
A wire basket or mesh bin			
Black card			
White card			
Something you choose — write it in			

WHAT YOU FOUND

The surface it could not see at all was _____.

Black card read _____ and white card read _____.

So does the **color** of something change what the sensor sees? _____

THINK ABOUT IT

The cushion and the angled board from worksheet 1 **both** gave no reading. But they failed for **different reasons**. What is the difference?

_____

_____

SO WHAT DO YOU ACTUALLY DO ABOUT IT?

When the surface is angled

You found two ways to make the sensor fail. Knowing *why* is only half of it — **an engineer has to decide what to do.** Here is the first one.

Keep it square

If the reading starts climbing, you are probably **crooked**, not further away — the beam is taking a longer path to the same wall. **Check the sensor is square before every run**, and if your robot can steer, correct the **heading** and not just the distance.

Sweep, do not stare

Turn the robot **by hand** a few degrees each way, reading at each step. One fixed reading is **one chance** to catch the echo. A sweep gives you several, and one will be close to square on.

Treat “nothing” as information

No reading does **not** mean empty space. It means empty space *or* something angled *or* something soft. Never write **if no reading → drive forward**.

Use contact for the last few centimeters

Below 3 cm the sensor is blind anyway. If your plan needs to know it has arrived, that has to come from touching, not from looking.

WHICH OF THESE FOUR WOULD STILL WORK

… if the angled thing was **moving**, and you could not choose where it was pointing?

SO WHAT DO YOU ACTUALLY DO ABOUT IT?

When the surface is soft

WHY THIS ONE IS DIFFERENT

Soft means **mostly air, held in fibres** — which is very nearly what the air around it already is. So there is barely a boundary there for the click to bounce off. It goes in instead, and the fibres turn it into heat.

This is not a weak sensor. Being soft and being invisible to sound are almost the same thing. More power will not fix it. Neither will better code.

Detect it by the echo it steals

The one answer that needs **no new hardware at all**. If a wall behind it normally reads 80 cm and suddenly reads nothing, something that absorbs sound is in the way. You found the invisible thing **by what it took away.**

Change the energy: light

Light reflects off the surface of the fibres, and it does not care how much air is in between. An infrared sensor sees a cushion at up to about **70 cm**. The color sensor also uses light, but only works within a centimeter or two — by then you have already hit it.

Change the energy: touch

A touch sensor works on **anything that can push back**, whatever it is made of. The catch is obvious: it tells you at the moment of contact, and not one centimeter sooner.

WHY THIS MATTERS ON DECEMBER 9

Your opponent is a LEGO robot — **hard**, so it sends the click back, but covered in **angles, gaps and holes**. You will see it, lose it, and see it again.

So do not build a plan that has to watch it the whole time. **Build one that survives the robot vanishing for a second.**

THE RULE UNDERNEATH ALL OF THIS

Every sensor relies on **one** property of the thing it is looking at. Sound needs something hard. Light needs something that reflects. Touch needs contact. **Anything without that property is not badly seen — it is not seen at all.**

That is why real robots carry several. Not spares. **Different blind spots.**